Machine Learning Based Road Accident Severity Prediction Using Environmental and Traffic Features

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***Abstract*—Road traffic accidents are one of the major public safety problems in the world, claiming millions of lives each year. Kills thousands of people and injures thousands yearly, resulting in significant economic damage. A precise prediction of the severity of accidents is essential to These provide use for real-time emergency response, road safety policy, and intelligent transportation systems. A comparative study of machine learning (ML) algorithms – Naive Bayes, Logistic Regression, Multilayer Perceptron (MLP), and Support Vector Machine (SVM) is presented on paper. Sourced: Back propagation (BP) and Perceptron (MLP) Neural Network, and Random Forest—for predicting accident severity based on environmental and human behavior characteristics. traffic features. A real-world accident data set is preprocessed by imputing missing values in this data set and encoding the labels. Categorical variables and standard feature scaling. Training and evaluation of models is done by accuracy, precision, and recall. F1 score, and area under the ROC curve (AUC). The experimental results show that the Random Forest (RF) classifier achieves better predictive performance. Identifies weather conditions, road type, vehicle type, and the road as features that most significantly influence the speed. Named type and time-related attributes are the most important ones influencing the severity of the accidents.**

***Index Terms*—road accident severity prediction, machine learning, random forest, logistic regression, neural network, naive bayes, traffic safety, environmental features.**

I. Introduction

Road traffic crashes are one of the most critical public health and safety issues of the modern world. According to the World Health Organization (WHO), approximately 1.35 million people die every year as a result of road traffic crashes, with tens of millions more suffering non-fatal injuries. In addition to the personal impact, road crashes have high economic costs on countries, including health costs, loss of productivity, and infrastructure losses.

The degree to which a traffic accident is serious depends on a variety of factors, such as driver behavior, vehicle attributes, road conditions, and more. Infrastructure, weather/lighting, and traffic patterns. Traditional statistical models—such as Historically, these models (logistic regression, Poisson regression, negative binomial models, and ordered probit models) have been used to model them. Analyse accident information and determine contributing factors. However, these methods do not usually incorporate the nonlinear and high-dimensional They find relationships that exist in real-life accident data sets [11] and [12].

The recent progress of machine learning (ML) and artificial intelligence (AI) has led to a new array of opportunities for more accurate and interpretable ML model accident severity prediction. ML algorithms such as decision tree, random forest, support vector machine, and deep neural networks In modeling complex patterns within networks, they have shown better results than traditional statistical methods have been able to achieve. large-scale datasets [1], [2], [5].

This study is one of a set of studies that adopt and compare four well established machine in an organized and systematic approach. Training classifiers: Naive Bayes, Logistic Regression, Multilayer Perceptron (MLP), and Random Forest on an accident dataset. Enriched with environmental features and traffic features. The main goals are: (i) to preprocess and prepare an accident dataset for (ii) to extract and identify relevant features, (iii) to extract the relevant features and their associated values, and (iv) to store them in a database. To train and test various machine learning models to predict the severity of an accident; (iii) to compare model performance across several ML models; (iv) to demonstrate how ML can aid in traffic management. To measure the performance with standard classification metrics, and (iv) to determine which of the most influential features is responsible for the level of severity of an accident.

II. Related Work

Road traffic accidents are a significant public safety problem with many millions of fatalities, injuries, and economic losses occurring annually worldwide. As transportation complexity

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increases, analytical tools have been created to better analyse accident patterns. Traditional statistical models, like logistic regression, Poisson regression, and ordered probit models, have been extensively used but have limitations in their ability to describe the complex, nonlinear relationships between various factors such as driver behavior, vehicle condition, road infrastructure, and environmental conditions. For this reason, machine learning (ML) techniques have become a viable approach to process large scale data and to enhance predictions [11, 12].

In the initial studies, supervised ML algorithms were used to predict the severity of the accident, including Decision Tree, Random Forest, Multilayer Perceptron (MLP), and Naive Bayes. Researches indicated that Random Forest model was superior to the others because it was able to give more accurate prediction and have a capacity to handle complex data [1] and [2]. In the course of time, more sophisticated methods like Support Vector Machine (SVM), Artificial Neural Network (ANN) and ensemble methods have been adopted which have proved their ability in establishing relationships between variables such as location, road geometry, speed and environmental conditions [1], [2], [5], [15].

The use of neural network models, such as MLP and deep learning models like Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), has also enhanced the predictive accuracy. An example of this is that LSTM-based RNN models have been found to outperform traditional and basic ML based models for accident severity prediction [11].

In recent years, there has also been a focus on model interpretability to aid in policy decisions. Methods like SHAP analysis can be used to find out important factors like time, vehicle type, location, and road factors. For example, in the accident severity prediction problem in Dhaka, LightGBM showed good performance (ROC-AUC 0.72), emphasizing the need for explainable ML models [3].

The effectiveness of ML is further confirmed with regional studies based on real-world data. These algorithms have shown to be very accurate for predicting the severity of the accident with the major factors being vehicle type, driver age, road conditions, and traffic conditions [4] [5] [6].

For better performance, hybrid and ensemble models have been proposed, which involve the use of various algorithms including MLP, RF and XGBoost. These models have reached high accuracy, showing the benefit of the combination of the various techniques [7, 10]. Furthermore, ML has been embedded in the Geographic Information Systems (GIS) to create spatial analysis and find zones with a high rate of accidents in such a way that it has a good predictive capability in spatial analysis [19], [20].

In conclusion, the literature suggests that models based on ML outperform their statistical counterparts in predicting the severity of accidents. Future research will focus on explainable artificial intelligence, real-time data integration, and intelligent transportation systems, to enhance road safety and to facilitate data-driven decision making [16]–[18].

III. Dataset Preprocessing

A. Description Of Dataset

The study employs a real-life set of road accidents with different attributes, including weather, Environmental factors, road characteristics, and traffic features. The target variable: Weather_y, which is the current weather conditions. The classification target for accident severity-related environmental prediction is the conditions at the time of each accident.

B. Data Analysis

The dimensionality of the dataset was initially explored by using the dimension function, ”shape” on the data frame,df. Numerical and categorical features were determined independently. Means, standard deviations, minimum and maximum values were calculated for numeric variables. Pearson correlation between numbers. Multicollinearity was evaluated for features by using a heatmap. Value counts of the target variable were examined to check the distribution of the target variable. To evaluate possible class imbalance.

C. Handling Missing Value

Records that had missing data values in the target variable Weather_y were excluded. Methods for remaining missing data included median imputation. applied to numerical columns (to reduce the influence of outlier values), and mode imputation was applied to categorical columns, following The accident data preprocessing method is similar to the one standard in the field [4, 6].

$$x_{\text{imputed}} = \begin{cases} \text{median}(x_j) & \text{if } x \text{ is numerical} \\ \text{mode}(x_j) & \text{if } x \text{ is categorical} \end{cases} \tag{1}$$

D. Feature Scaling & Encoding

Each categorical feature and the target were encoded with the scikit-learn LabelEncoder that assigns a unique number to each category. To an integer label. Stratified random splitting with a 70/30 ratio was used to split the dataset into training and testing subsets. Set the random seed (random_state=42). All the feature values were normalized using the StandardScaler to scale the data into zero mean and unit variance.
Calculated using the training set and used throughout the test set:

$$x' = \frac{x - \mu}{\sigma} \tag{2}$$

where μ is the mean and σ is the standard deviation of each feature, computed on the training set and applied consistently to the test set. Feature scaling is particularly important for Logistic Regression and MLP, which are sensitive to feature magnitude.

For Logistic Regression and MLP, which are sensitive to the feature values' magnitude, feature scaling is especially crucial.

IV. Methodology

A. Machine Learning Models

*1) **Naive Bayes (GaussianNB)**:* A probabilistic classifier based on Bayes's Theorem with the assumption of conditional independence between features. The Gaussian variant model assumes that each feature is modeled as a continuous Gaussian distribution and has a low computational cost. It serves as a fast probabilistic baseline.

*2) **Logistic Regression**:* A linear discriminative classifier that models the posterior probability of each class with the softmax function in the multiclass case. Optimized using L-BFGS, max_iter=1000. Computes interpretable coefficient weights and a competitive linear baseline [5].

*3) **Multilayer Perceptron (MLP)**:* A feedforward artificial neural network with one hidden layer with 100 neurons and ReLU activation. This model was trained for 1000 iterations using the backpropagation algorithm with the Adam optimizer. The MLP is capable of modeling nonlinear feature interactions that linear models cannot. As in previous studies of accident severity [1, 20], models cannot represent.

*4) **Random Forest**:* Bootstrap samples, random feature subsets, and n_estimators=100, meaning that there are 100 decision trees.Ensemble of 100 decision trees (n_estimators=100) trained on bootstrap samples with random feature subsets at each split. The majority voting method is used to predict. Random Forest is a powerful algorithm that is resistant to overfitting and has the natural ability to offer feature importance. Scores are typically computed by mean decrease in Gini impurity, and are thus commonly used in research in accident severity [5], [6], [15], [16].

B. Evaluation Metrics

All models were compared by the following measures: **Accuracy** (proportion of correctly classified instances); **Weighted Precision** (mean of the precision over all classes); **Recall** (proportion of correct classifications across all classes); and **F1 score** (weighted harmonic mean of precision and recall). weighted by class support); **Weighted F1-Score** (average of/with class support); **Weighted-g-mean** (geometric mean of class supports); **Weighted Log-likelihood** (log-likelihood divided by class support); Precision and recall); and **AUC-ROC** (area under the One-vs-Rest ROC curve, averaged across all classes). These metrics collectively make a complete picture of model performances, taking into account class imbalance by weighted averaging.

C. Experimental Pipeline

The complete experimental workflow is illustrated as follows:

1) Load raw dataset from CSV file.
2) Perform exploratory data analysis (EDA) and correlation analysis.
3) Handle missing values via median/mode imputation.
4) Encode categorical variables with Label Encoding.
5) Split data into training (70%) and test (30%) sets.
6) Apply StandardScaler normalization.
7) Train all four classifiers on scaled training data.
8) Evaluate each model on the held-out test set.
9) Visualize confusion matrices, ROC curves, and metric comparisons.
10) Analyze feature importances from the Random Forest model.

V. Results and Discussion

A. Model Performance Comparison

The table I shows the performance of all four classifiers on the test set. The highest overall classification rate is obtained by the Random Forest classifier. The results for the five metrics are presented with scores, which are consistent with results reported in the traffic safety ML literature [5], [6], [15], and [16].
Table I summarizes the performance of all four classifiers evaluated on the test set. Results are reported for accuracy, weighted precision, weighted recall, weighted F1-score, and AUC-ROC.

TABLE I
Comparative Performance of Machine Learning Models

Model	Acc.	Prec.	Rec.	F1	AUC
Naive Bayes	0.1153	0.4181	0.1153	0.0591	0.5460
Logistic Regression	0.3997	0.3044	0.3997	0.3190	0.5498
Neural Network (MLP)	0.4595	0.4403	0.4595	0.4189	0.6511
Random Forest	0.7855	0.8057	0.7855	0.7814	0.9268

Note: Values to be populated with experimental outputs from the trained models.

B. Discussion

The Random Forest classifier is the one that is likely to perform best on all measures. This is due to:
Several strengths of ensemble tree-based models: (i) robustness due to averaging many diverse trees, which reduces (iii) ability to learn complex non-linear relationships

of features via random subspace selection; (iv) resistance to overfitting; and (v) ability to learn complex patterns.(iv) resistance to overfitting, (v) ability to learn complex patterns, and (i) ability to capture complex nonlinear relationships of features via random subspace selection. and (iii) stability through bootstrap aggregation (bagging) with the aim of better generalizing to new data. These results are consistent with previous studies. An Obasi and Benson [5] and Mengistu et al. [6] observed that RF was the best model in all their studies, Yang et al. [15] reported the same, and Jaji [16] got RF as the best performer in one of his studies. All these studies reported RF as the best model, as did Obasi and Benson [5], Mengistu et al. [6], Yang et al. [15], and Jaji [16] in one of his studies. Tasks for the classification of accident severity.

The MLP Neural Network is 2nd, because of its ability to model non-linear relationships. Logistic Regression provides a Competitive linear baseline and interpretable coefficients. Naive Bayes is a simple, fast algorithm, but it can be inaccurate because of its strong conditional independence assumption, often not met in accident data that contain correlated features.

C. *Feature Importance Analysis*

The feature importance scores obtained from the Random Forest modeling show those predictors that are most influential in determining the severity of the accidents. Attributes related to the most likely features to appear will be against weather conditions, road type, car category, time of day, and junction type. This is consistent. Secondly, there is no consensus about which are the most important factors [4, 12]. Labib et al. [4] identify vehicle type, road geometry, traffic control, and weather as the most significant factors, while with the Casualty class, accident time, and road type were among the major drivers of severity, which were identified by Rifat et al.[3].

D. *ROC Curve Analysis*

Multi-class ROC curves (One-vs-Rest) verify the discriminative power of each one in all severity classes. The Random AUC-ROC values of the forest model are consistently high and indicate its superiority, which is in line with the results obtained by Mengistu et al. [6] (AUC-ROC = 0.87) and Rifat et al. [3] (AUC = 0.72 for LightGBM). The confusion matrix analysis also shows that Random Forest has the least number of off-diagonal elements. Misclassifications in each severity class.

VI. Conclusion

This paper introduced a machine learning model to predict the severity of road accidents based on environmental and traffic features. Four classifiers (Naive Bayes, Logistic Regression, Multilayer Perceptron, Random Forest) were systematically trained, and their performance was evaluated. Evaluated and compared on a real-world accident data set. The pipeline used for the experiment consisted of detailed processing (data missing) and other components. value imputation, label encoding, feature scaling, evaluation on five standard metrics, and visualization using confusion matrices, Rockies and feathers. It is shown that the overall prediction accuracy and precision of the Random Forest ensemble classifier are highest, as well as the recall, F1 score, and reporting and recall rate. The results showed that all the evaluated models performed better than the AUC, and no other models performed consistently better across all traffic safety evaluations. literature. Weather conditions, road type, vehicle type, and time-related variables are the most important features, as determined by feature importance analysis. factors that are critical for accident severity. In practice, the use of severity prediction tools based on machine learning can help traffic management authorities in making decisions about the allocation of resources for emergency response. More efficiently, recognising high-risk environmental conditions, creating targeted road safety interventions, and providing information to the public. transportation system (ITS) development.

Future Work: Directions involve bringing in spatial GIS analysis to find out the accident hot spots [19] and using advanced deep learning techniques to understand the impact of the event on the model. using an attention mechanism [15] or transductive learning [7]; significantly improving accuracy by using an attention mechanism [15] or transductive learning [7]; adding SHAP or LIME for model explainability [3]; handling class imbalance by swapping out the loss function [7] or using an attention mechanism [15]. SMOTE [6] and extending this to real-time accident prediction with streaming traffic sensor data [9], [18].

Appendix

The complete Python implementation is provided below.

```
# Imports
import pandas as pd
import numpy as np
import the train_test_split function from the sklea
For Scaling, we use StandardScaler from sklearn.pre
GaussianNB is a type of Gaussian Naive Bayes.Gauss
import sklearn.linear_model as LR
import sklearn.neural_network as MLPClassifier
Import the RandomForestClassifier from the sklearn
from sklearn.metrics import (accuracy_score, preci
    recall_score, f1_score, confusion_matrix,
    classification_report, roc_auc_score, roc_curve
Import matplotlib.pyplot and set it as plt.Import n
Import the package seaborn.
```

```
# Load Dataset
df = pd.read_csv('Modified_Accident_Dataset_finaly.csv',
                 encoding='latin1')

# EDA
numerical_data = df.select_dtypes(include='number')
categorical_data = df.select_dtypes(include='object')

plt.figure(figsize=(12,8))
sns.heatmap(df.corr(), cmap='coolwarm', linewidths=0.5)
plt.title("Feature Correlation Heatmap")
plt.savefig("correlation_heatmap.png", dpi=300,
            bbox_inches='tight')
plt.show()

# Missing Value Handling
df = df.dropna(subset=['Weather_y'])
num_cols = df.select_dtypes(
    include=['int64', 'float64']).columns
df[num_cols] = df[num_cols].fillna(df[num_cols].median())
cat_cols = df.select_dtypes(include=['object']).columns
For each of the categories in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])

# Encoding
le = LabelEncoder()
For each column cat in cat_cols:
    df[col] = le.fit_transform(df[col])
df['Weather_y'] = LabelEncoder().fit_transform(
    df['Weather_y'])

# Train-Test Split
X = df.drop('Weather_y', axis=1)
y = df['Weather_y']
Split the data into training and test sets.Divide the data into a training set and a test set.
    X, y, test_size=0.3, random_state=42)

# Feature Scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Model Definitions
models = {
    "Naive Bayes": GaussianNB(),
    "Logistic Regression": LogisticRegression(
        max_iter=1000),
    "Neural Network": MLPClassifier(
        hidden_layer_sizes=(100,),
        max_iter=1000, random_state=42),
    "Random Forest": RandomForestClassifier(
        n_estimators=100, random_state=42)
}

# Evaluation Function
A function to evaluate a model based on its parameters, predictions, and probabilities.
def evaluate_model(name, model, X_test, y_test):
    y_pred = model.predict(X_test)
    y_proba = model.predict_proba(X_test)
    accuracy = accuracy_score(y_test, y_pred)
    precision = precision_score(y_test, y_pred,
        average='weighted')
    recall = recall_score(y_test, y_pred,
        average='weighted')
    f1 = f1_score(y_test, y_pred, average='weighted')
    auc = roc_auc_score(y_test, y_proba,
        multi_class='ovr')
    print(f"\n{name} Results")
    print(f"Accuracy: {accuracy:.4f}")
    print(f"Precision: {precision:.4f}")
    print(f"Recall: {recall:.4f}")
    print(f"F1 Score: {f1:.4f}")
    print(f"AUC: {auc:.4f}")
    print(classification_report(y_test, y_pred))
    Your answer should contain: return accuracy, p
        y_pred, y_proba
Train and Evaluate All models.
results = []
predictions = {}
for name in models.items():
    model.fit(X_train_scaled, y_train)
    acc, prec, rec, f1, auc, y_pred, y_proba = \
        evaluate_model(name, model,
                       X_test_scaled, y_test)
    results.append([name, acc, prec, rec, f1, auc])
    predictions[name] = (y_pred, y_proba)

# Results Table
results_df = pd.DataFrame(results,
    columns=['Model','Accuracy','Precision',
             'Recall','F1','AUC'])
print(results_df.sort_values('Accuracy',
                             ascending=False))

# Accuracy Bar Plot
plt.figure(figsize=(8,5))
sns.barplot(x='Model', y='Accuracy', data=results_
plt.title("Model Accuracy Comparison")
plt.ylim(0,1); plt.xticks(rotation=20)
plt.savefig("accuracy.png"); plt.show()

#(Precision-Recall Plots
plt.figure(figsize=(12,5))
plt.subplot(1,2,1)
sns.barplot(x='Model', y='Precision',
            data=results_df)
plt.title("Precision")
plt.subplot(1,2,2)
sns.barplot(x='Model', y='Recall', data=results_df
plt.title("Recall")
plt.tight_layout()
plt.savefig("precision_recall.png"); plt.show()
```

```
# Confusion Matrices
plt.figure(figsize=(16,4))
for i, (name, (y_pred, _)) in \
        enumerate(predictions.items()):
    plt.subplot(1,4,i+1)
    cm = confusion_matrix(y_test, y_pred)
    sns.heatmap(cm, annot=True, fmt='d')
    plt.title(name)
plt.tight_layout()
plt.savefig("confusion_matrix.png"); plt.show()

# ROC Curves
plt.figure(figsize=(8,6))
for name, (y_pred, y_proba) in predictions.items():
    for i in np.unique(y_test):
        y_binary = (y_test == i).astype(int)
        fpr, tpr, _ = roc_curve(y_binary,
                                y_proba[:, i])
        plt.plot(fpr, tpr,
                 label=f"{name} Class {i}")
plt.plot([0,1],[0,1],'k--')
plt.xlabel("FPR"); plt.ylabel("TPR")
plt.title("ROC Curve"); plt.legend()
plt.savefig("roc_curve.png"); plt.show()

# Feature Importance (Random Forest)
rf = models["Random Forest"]
feature_importance = pd.Series(
    rf.feature_importances_, index=X.columns)
feature_importance.sort_values(
    ascending=False).head(10).plot(kind='bar')
plt.title("Top 10 Important Features")
plt.show()
```


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